CSE228: Advanced Computer Programming

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Submitted to:

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Airplane Flight Reservations

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- Introduction

This application is made to overcome the problem of data missing and data corruption that may occur during reserving flights manually. such problems cost a lot of money on the long run.

This application allows airport employees to register new passengers, edit passengers' data, find passengers' data, cancel a passenger's reservation, register new flights, edit their data, show lists of passengers for every flight, search for flights, and cancel flights without any data corruption.

Here we introduce what we, developers, have been through to get this application out to the light, and how the user can use it easily.

- Program Interface

Here we actually didn't develop a GUI, but if there was one it'll be very similar to the console application window. Just by clicking on the icon, a menu appears asking the user what task is needed to be done. One of the options in this menu is exiting the program by entering the specified number.

- Input and Output

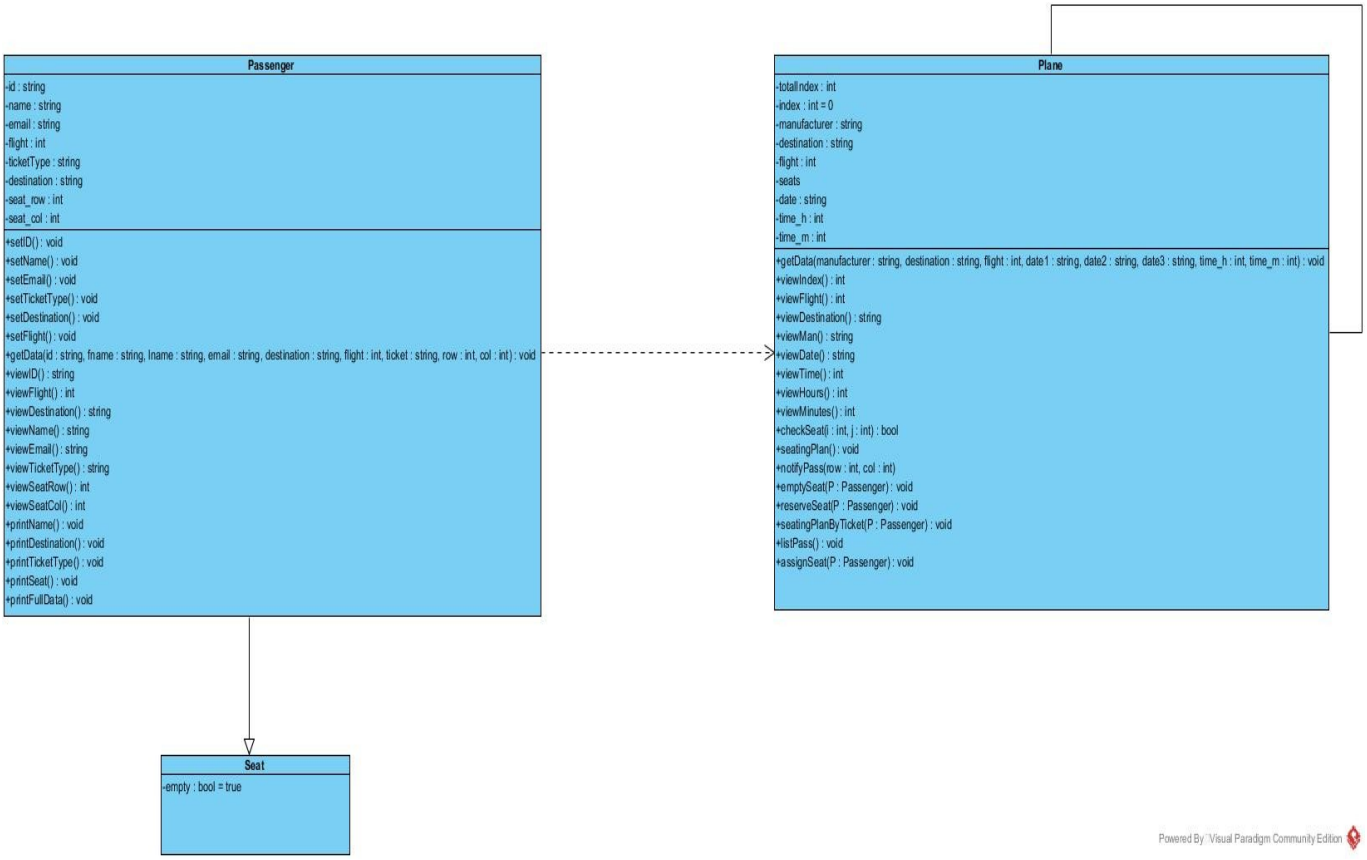
Format of input files:

FROM:	TO:	FLIGHT NO.	DAY	DATE	TIME
PARIS	CAIRO	128	SATURDAY	18TH AUGUST	9 : 15
BOEING	MADRID	140	SATURDAY	18TH AUGUST	12 : 25

Format of output files:

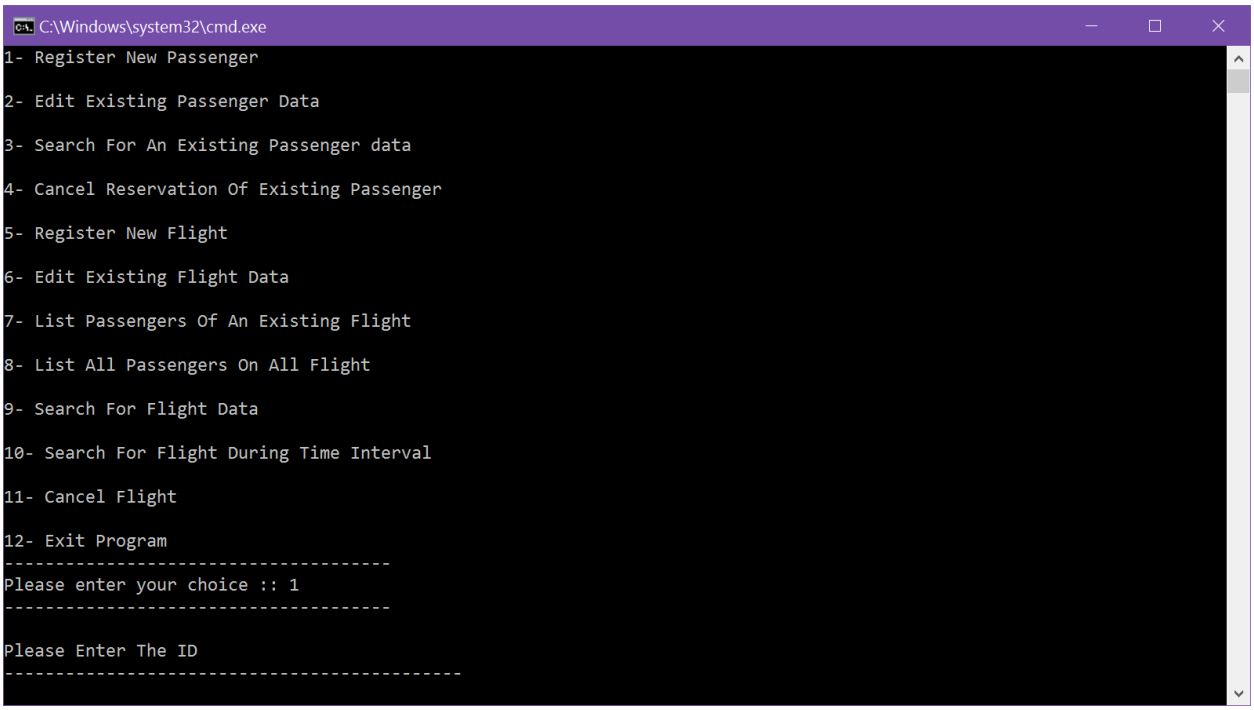
ID	NAME	EMAIL	DESTINATION	FLIGHT NO	CLASS	SEAT NO
P1001	SARA ADEL	SARA.ADEL@EXAMPLE.COM	PARIS	181	FIRST	1 1
P1002	KARIM NABIL	K.NABIL@EXAMPLE.COM	MADRID	140	FIRST	0 2

- Program Structure – technical manual



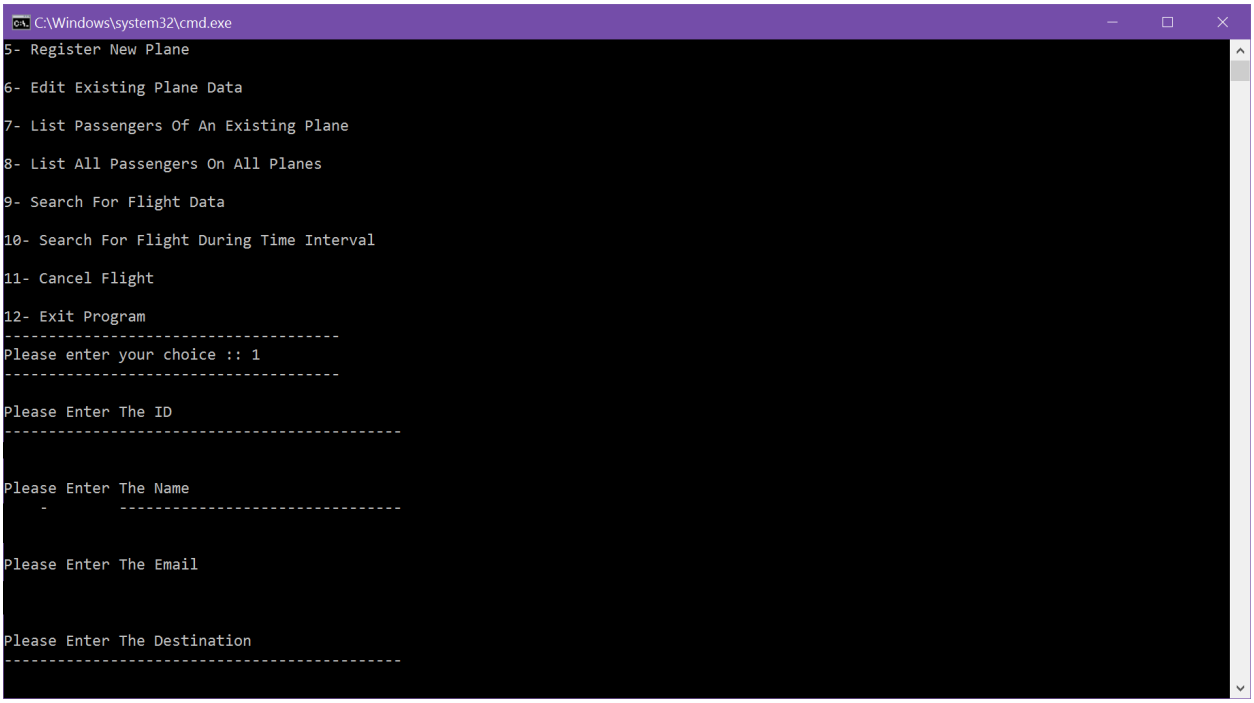
- Program Execution – user manual

Once the application is activated a menu appears asking the user to enter his/her choice.

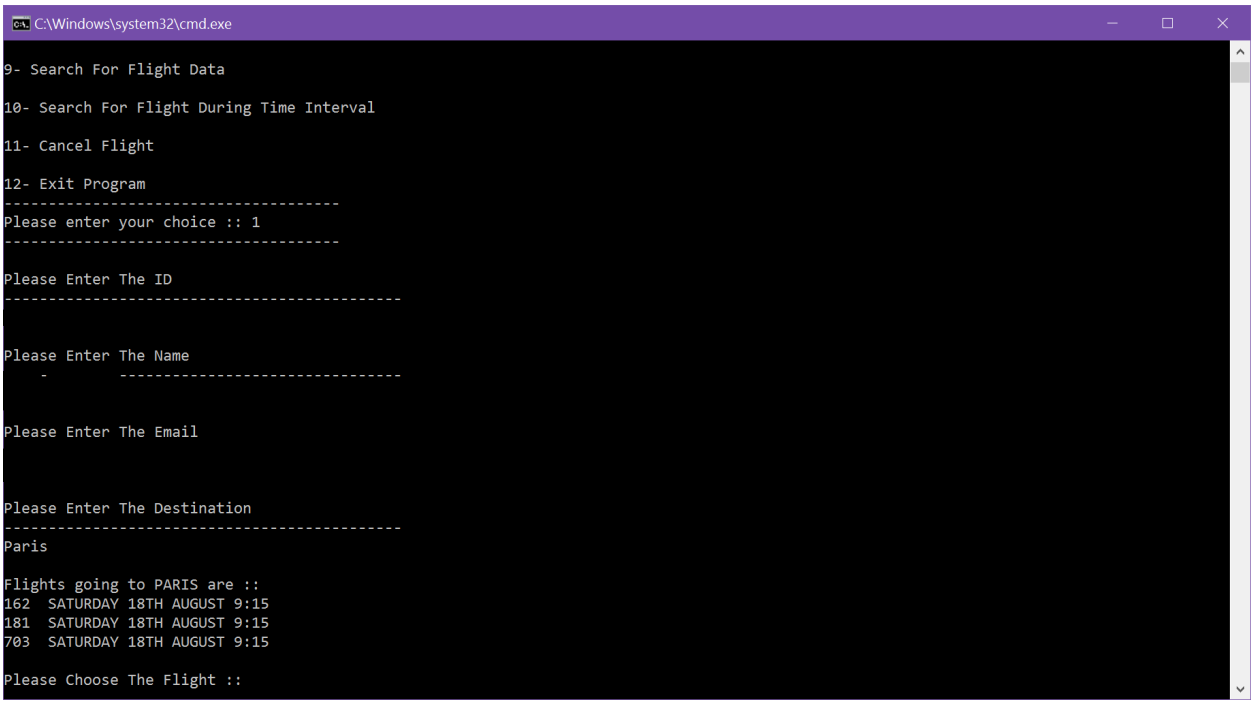


1. Registering new Passenger

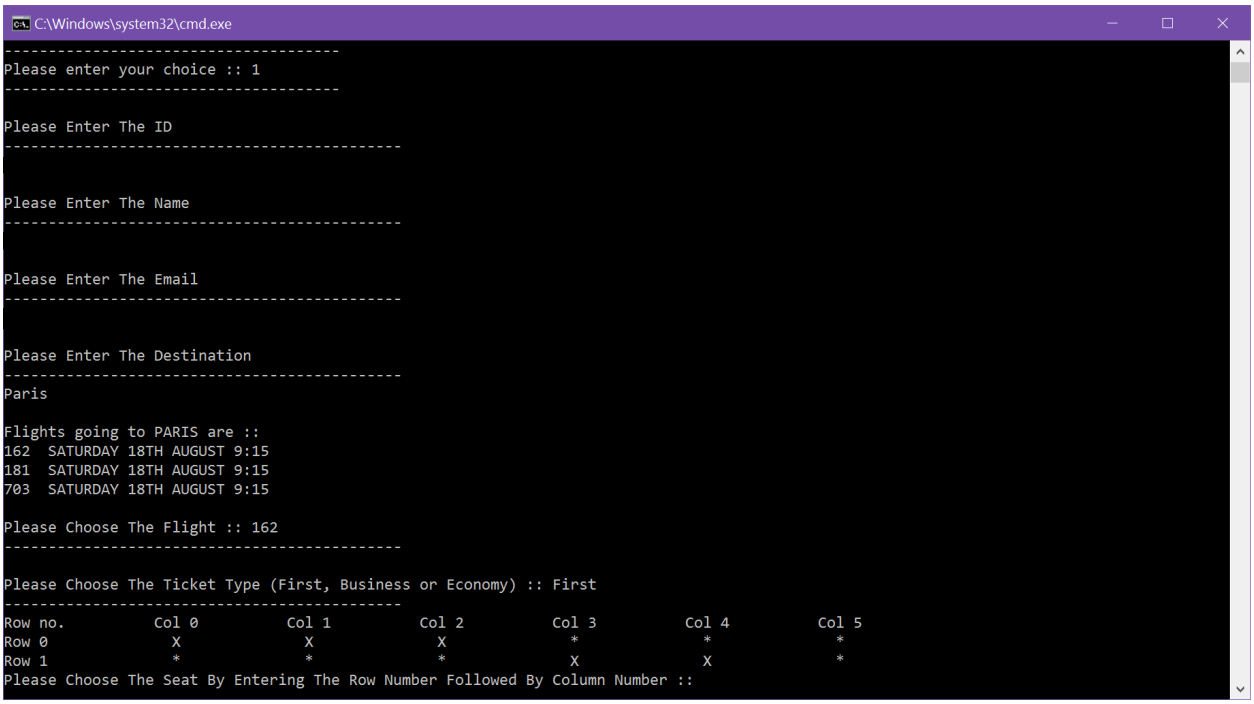
Enter passenger's ID, name, e-mail, destination.



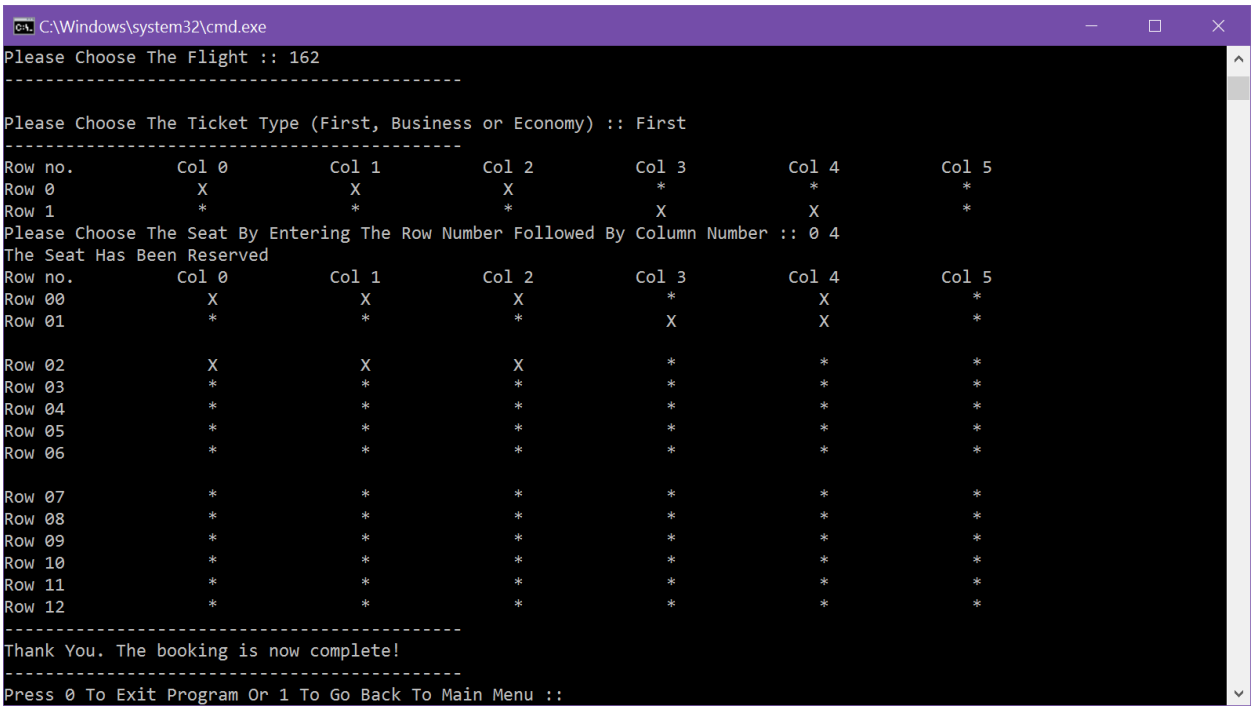
A list of flights still having vacancy will appear.



Enter desired flight number. Enter ticket type. Enter desired seat.

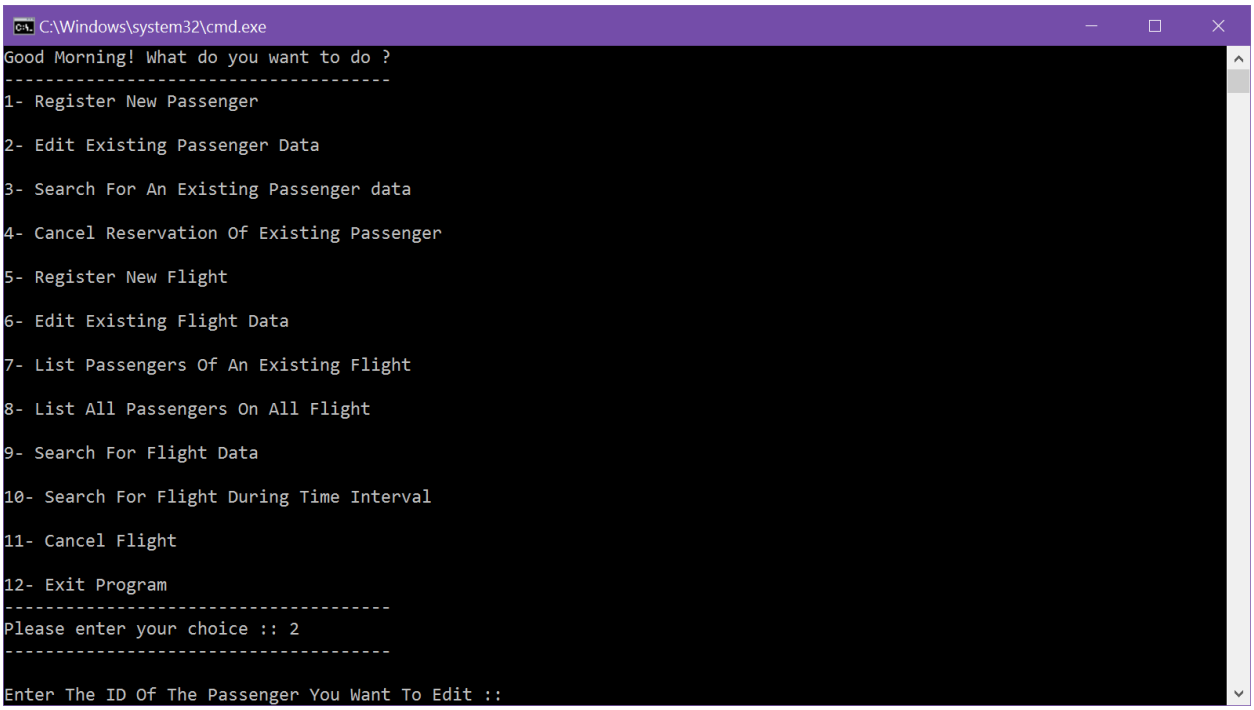


An announcement must appear to verify successful booking.

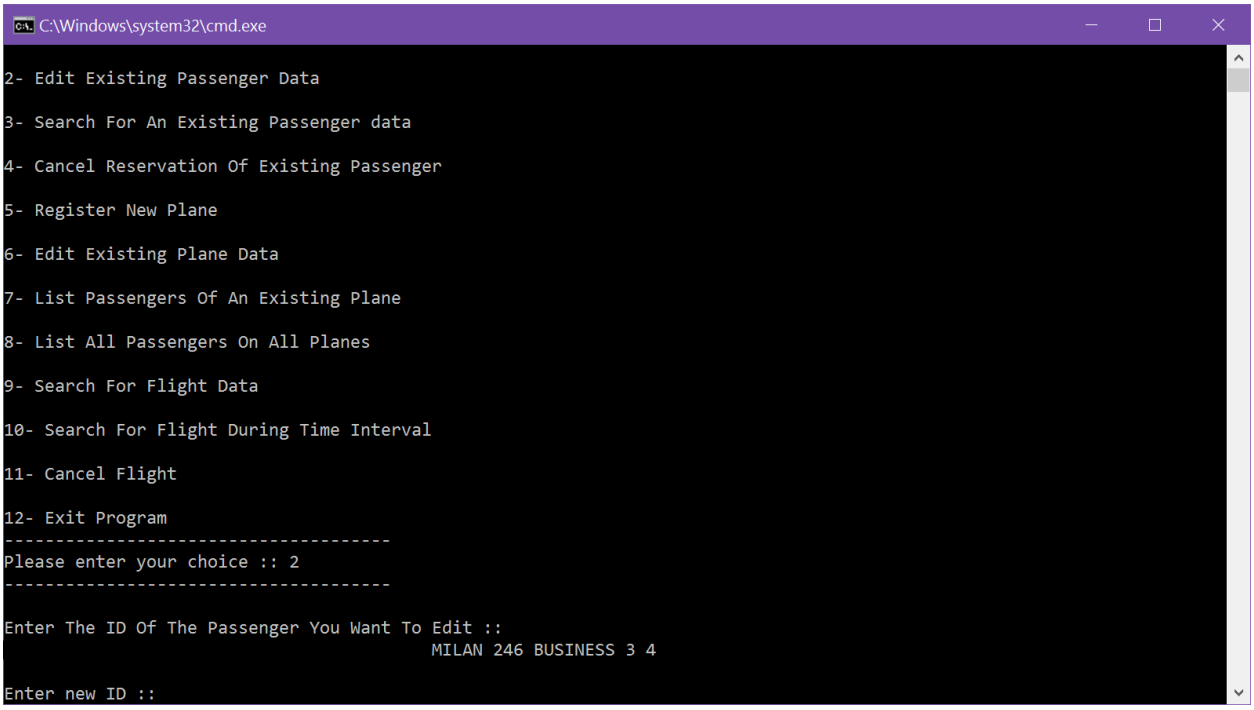


2. Edit existing passenger data

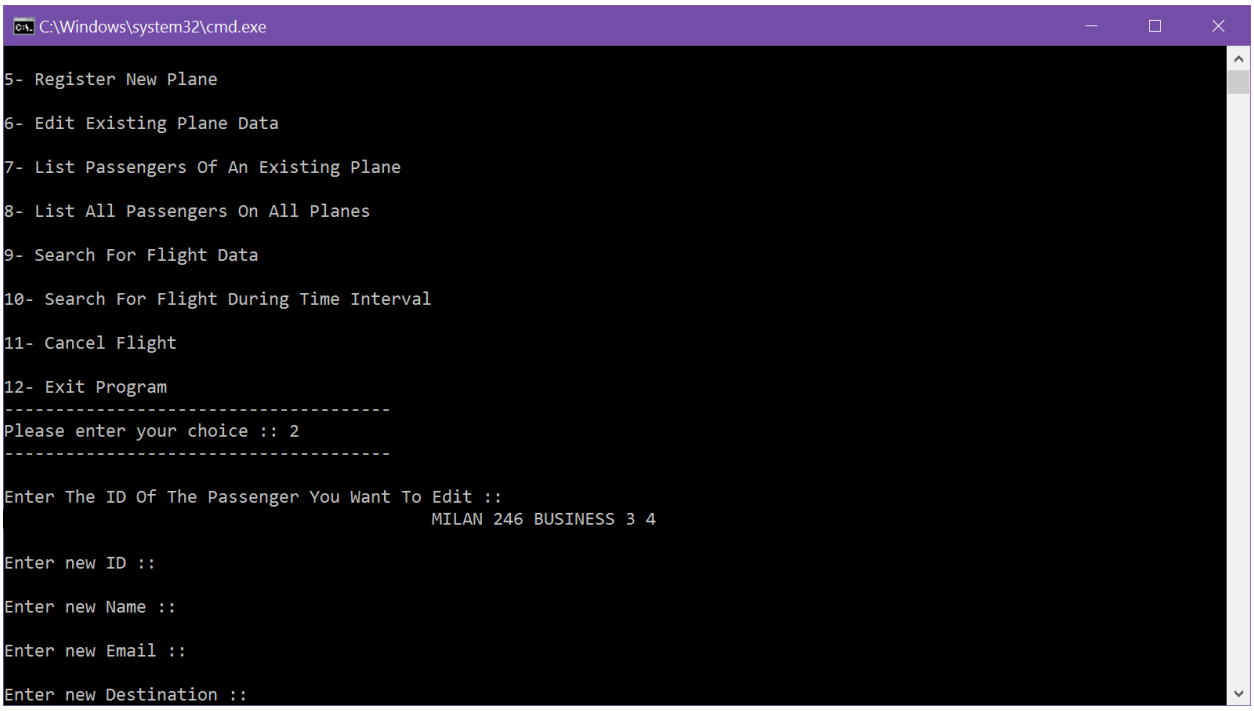
Enter ID passenger’s data requires editing.



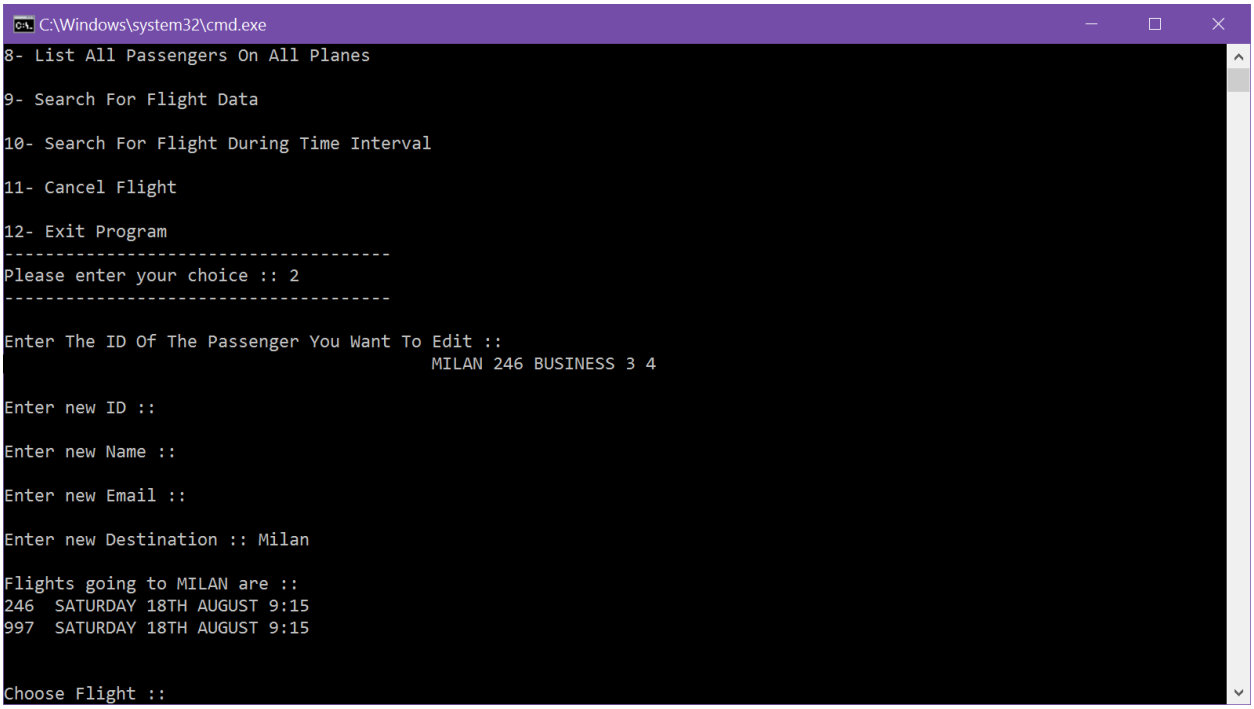
Passenger’s data appears.



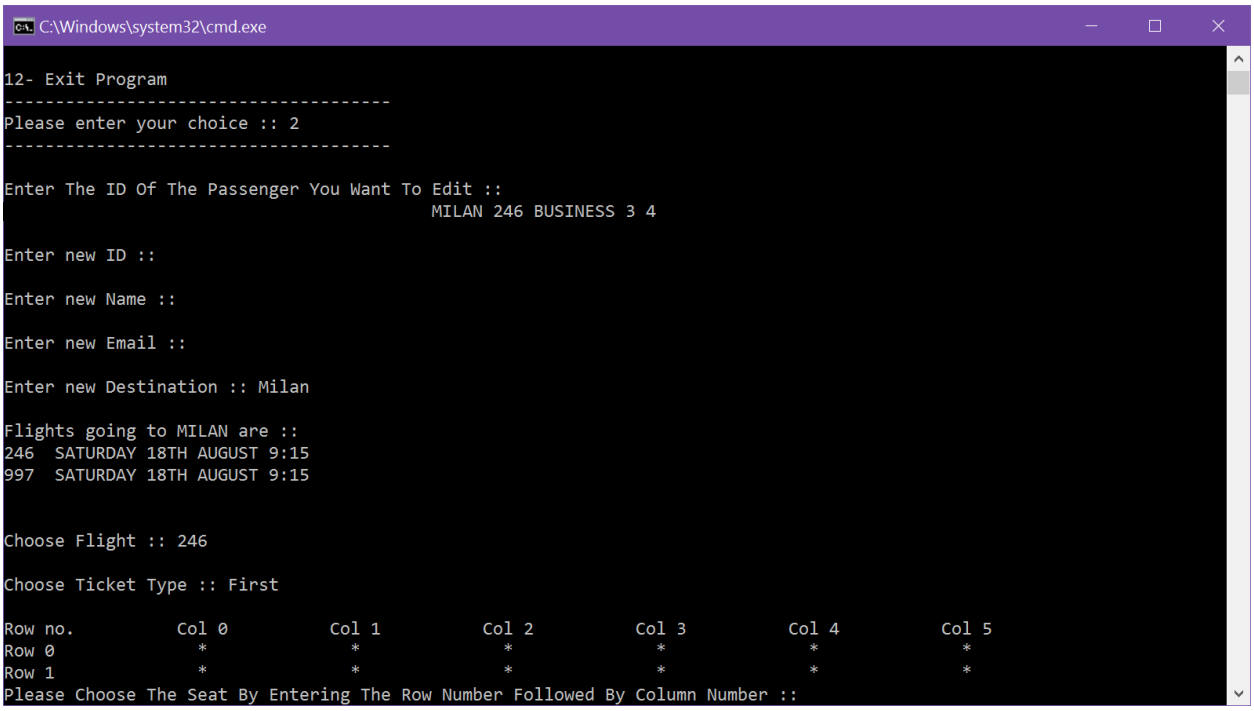
Re-enter new Passenger’s data; ID, name, e-mail, and destination.



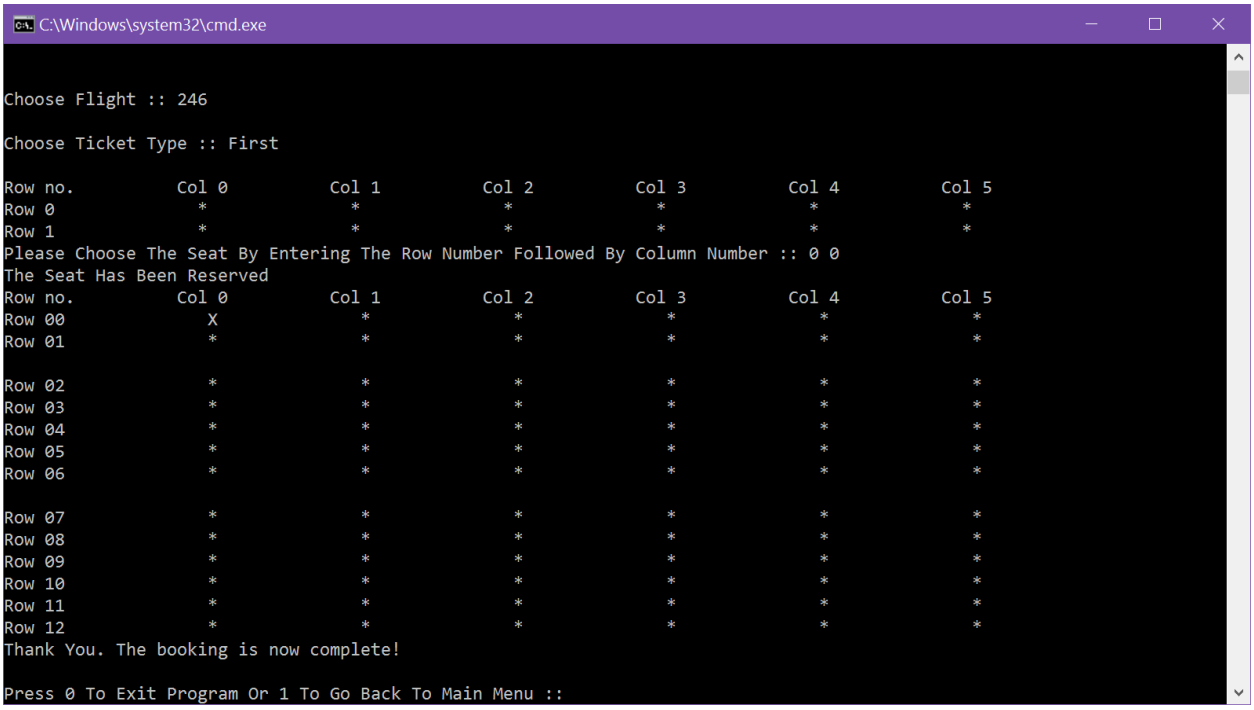
List of flights still having vacancy will appear.



Re-enter desired flight number. Enter ticket type. Enter desired seat.

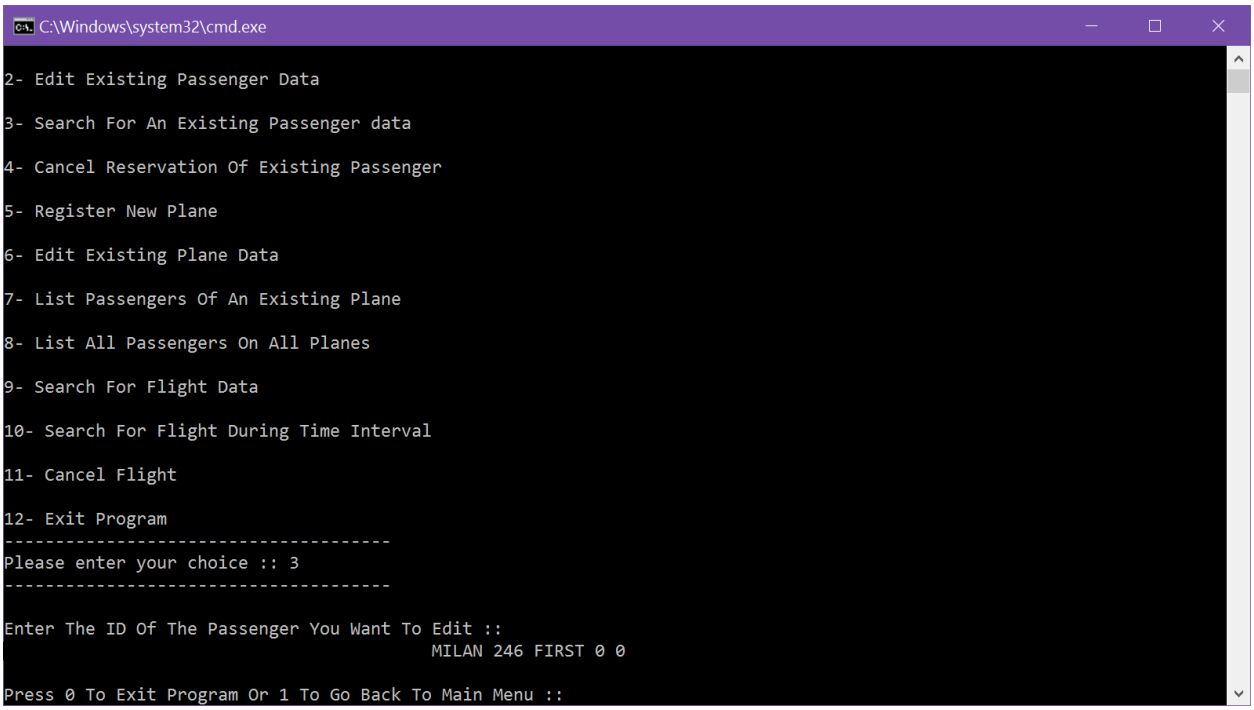


An announcement must appear to verify successful updating.



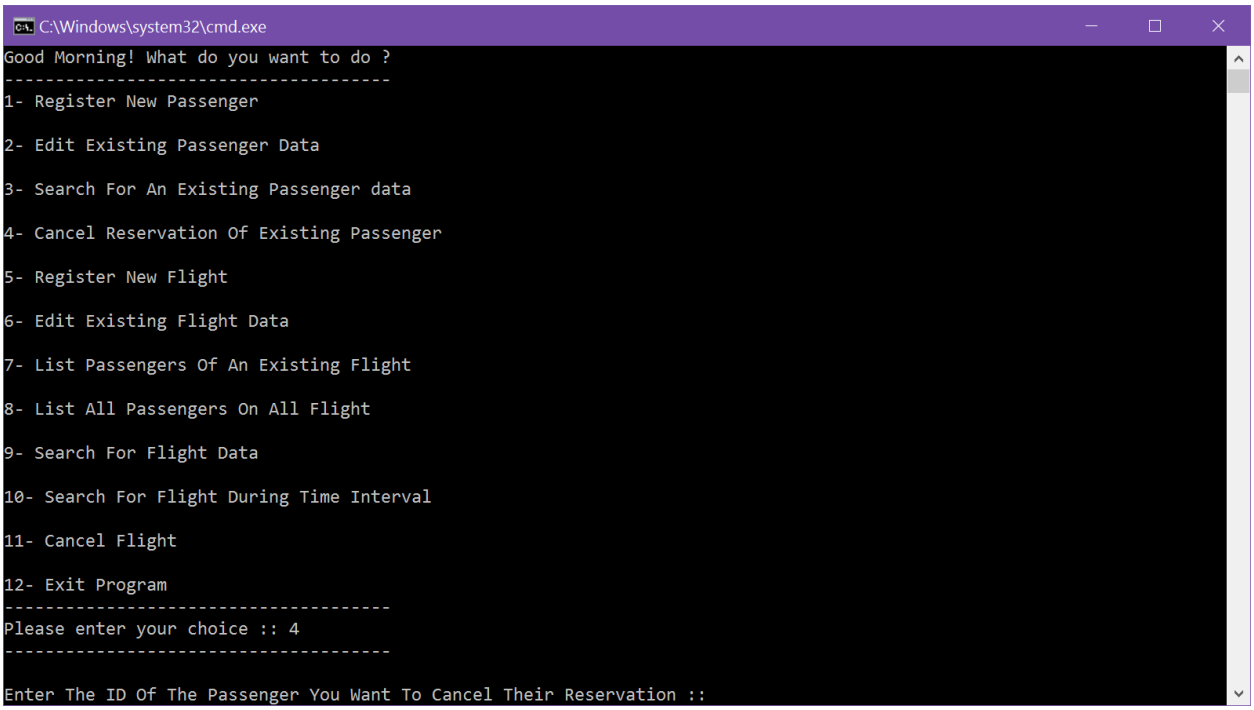
3. Search for an Existing Passenger data

Enter passenger’s ID required to search for. Passenger’s data will appear.

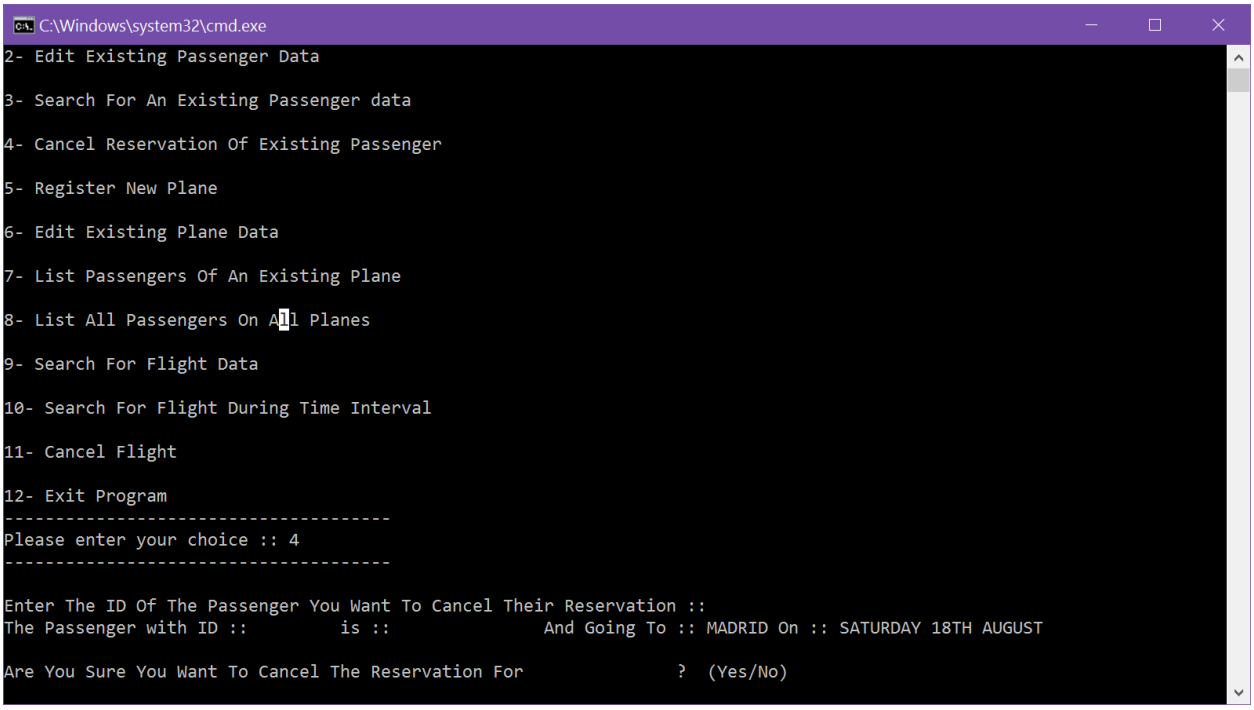


4. Cancel a passenger reservation

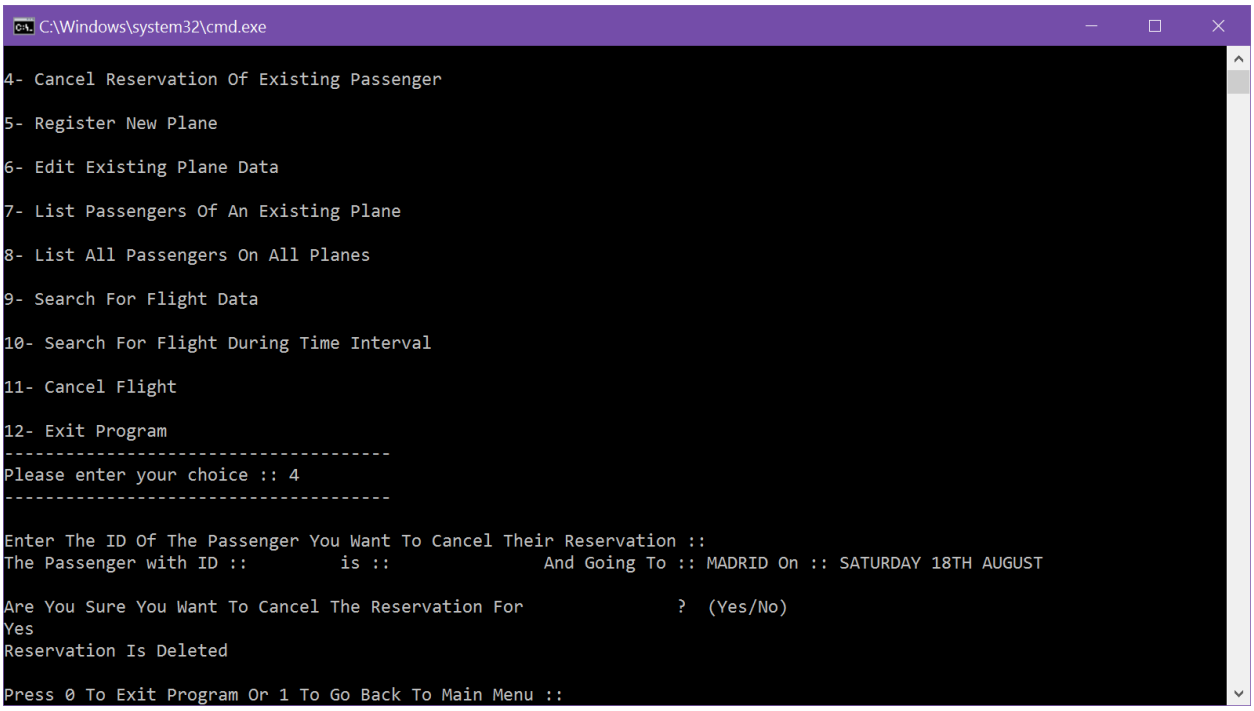
Enter passenger's ID required to cancel his/her reservation.



Type "Yes" if you're sure of cancelling this reservation.

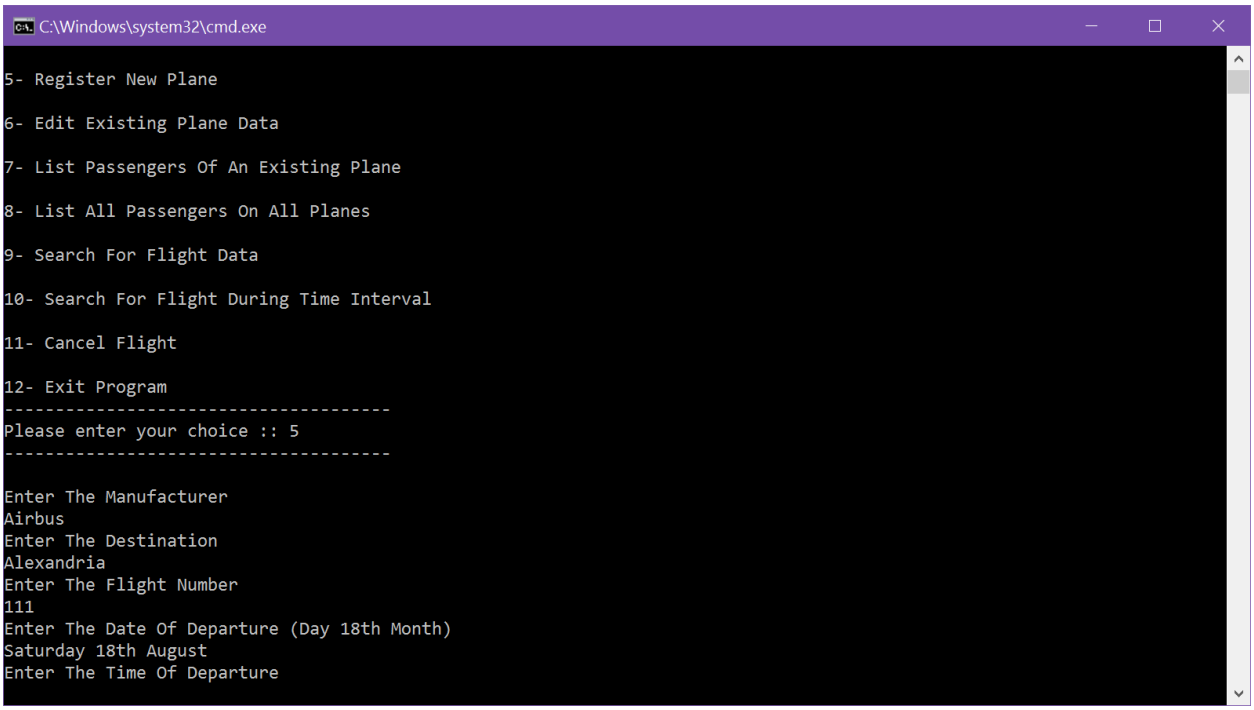


An announcement must appear to verify cancellation.

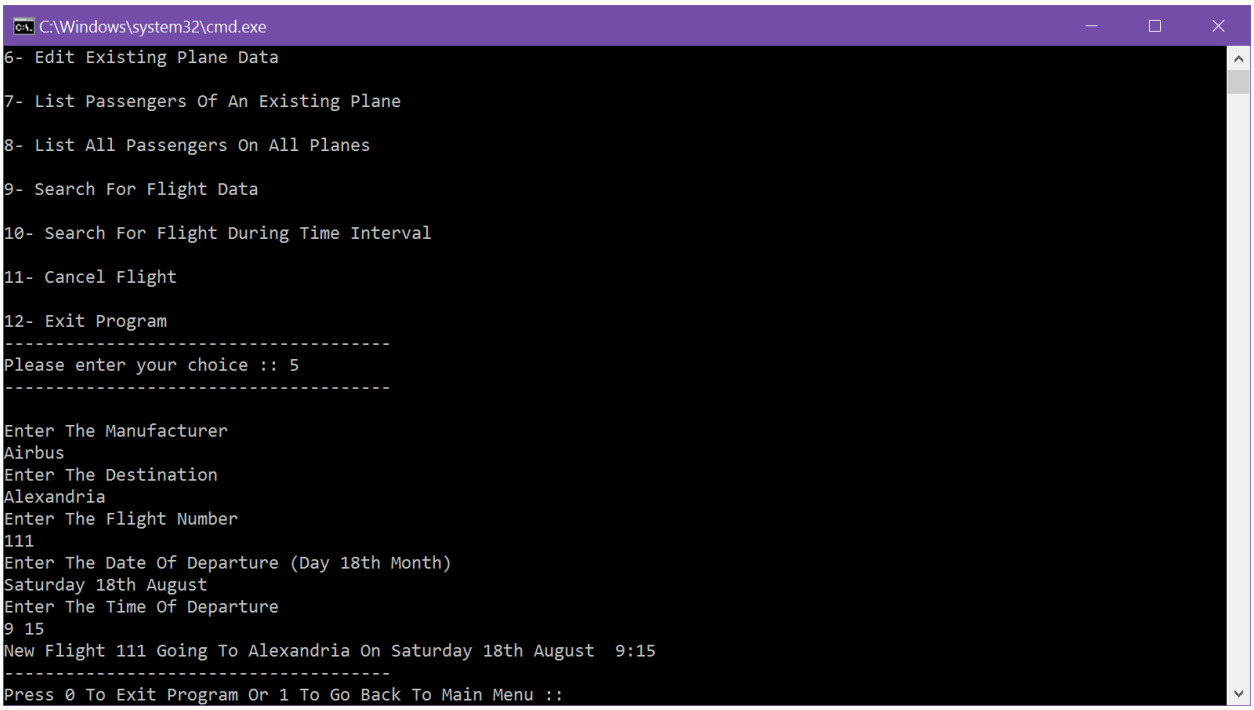


5. Register new flight

Enter flight's manufacturer, destination, and flight number. Enter date of departure, time of departure.

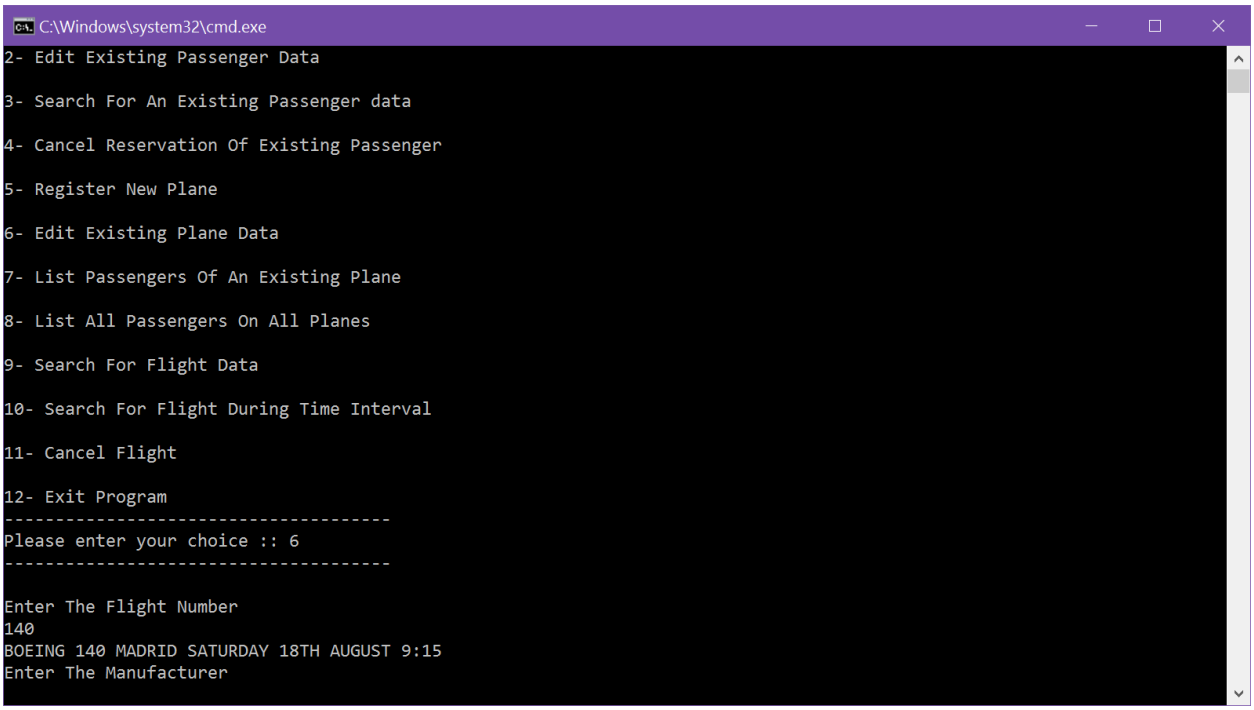


Data of new flight registered will appear.

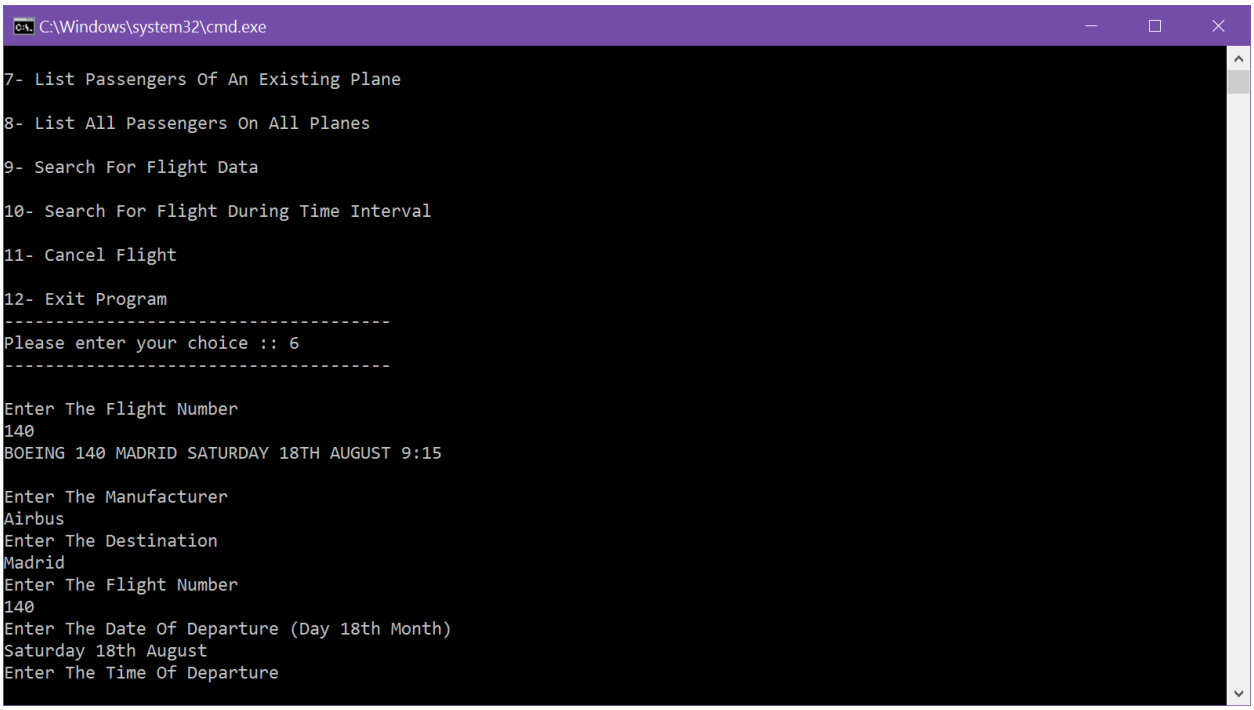


6. Edit Existing Flight data

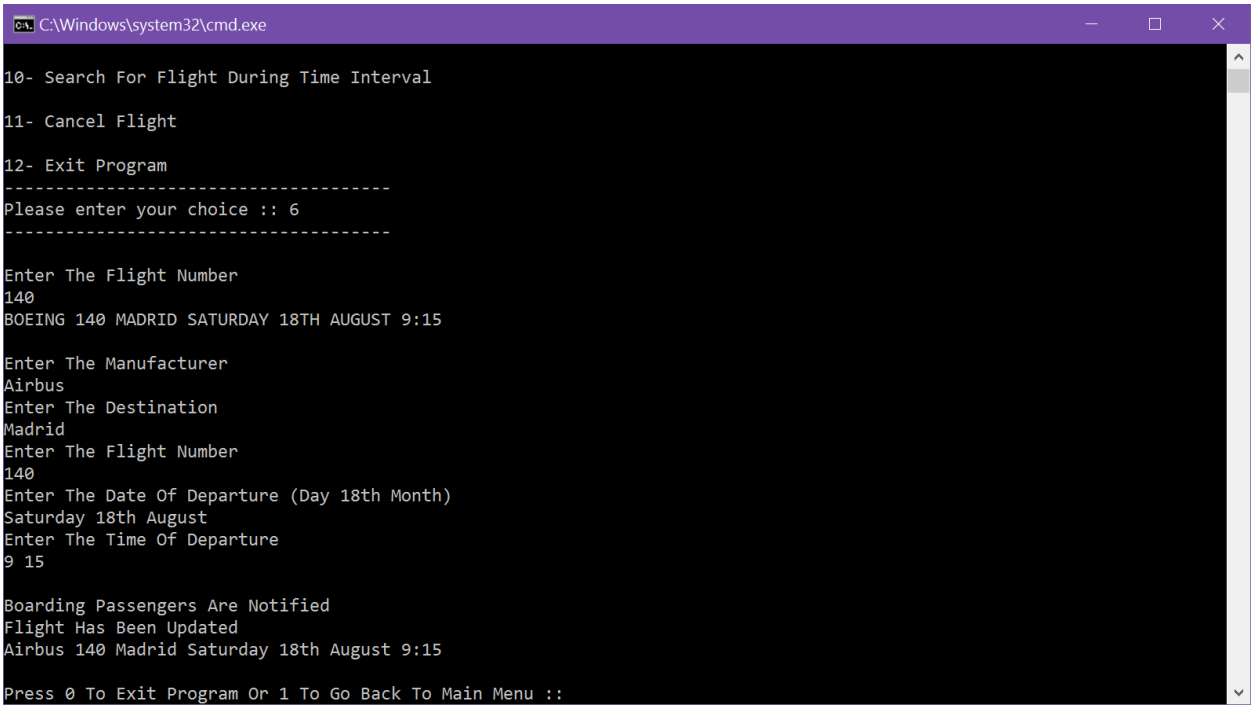
Enter flight number and its data will be shown.



Re-enter plane's manufacturer, destination, and flight number. Re-enter date of departure, time of departure.

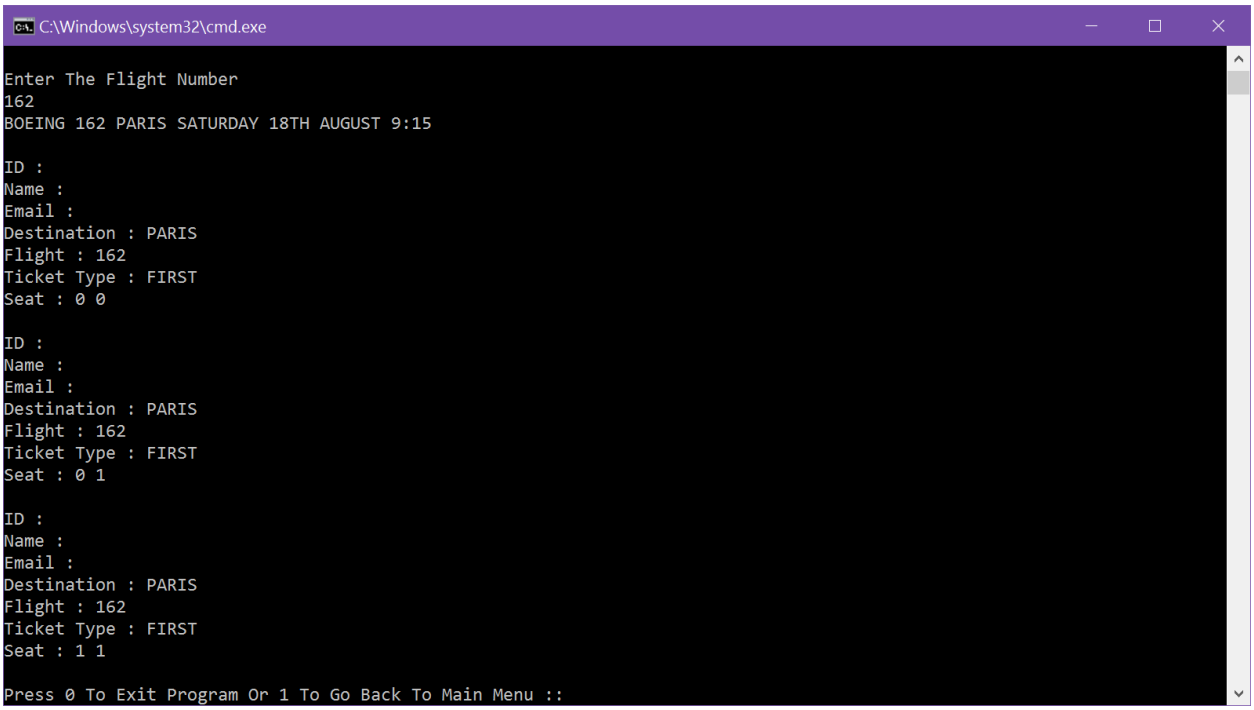


A message will be sent to passengers informing them of the new updates. An announcement must appear to verify successful updating. Flights data will appear.

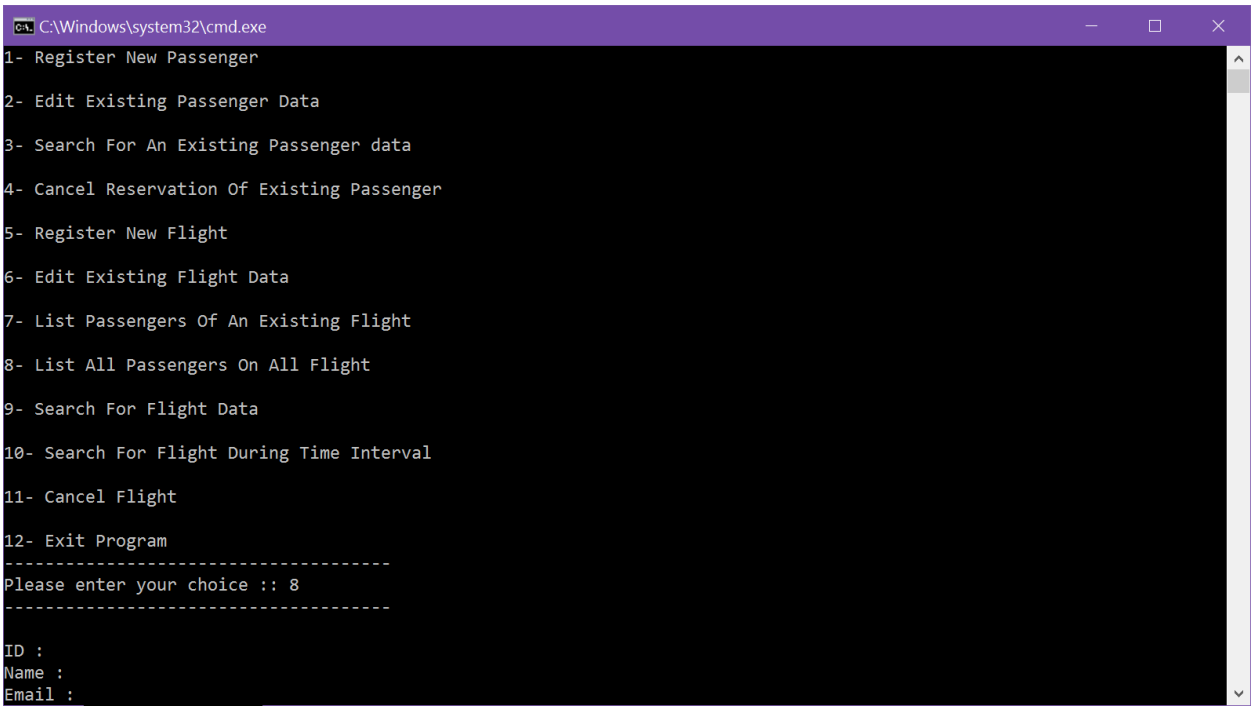


7. List of passengers of a flight

Enter flight number. All passengers' data and flight data will appear.

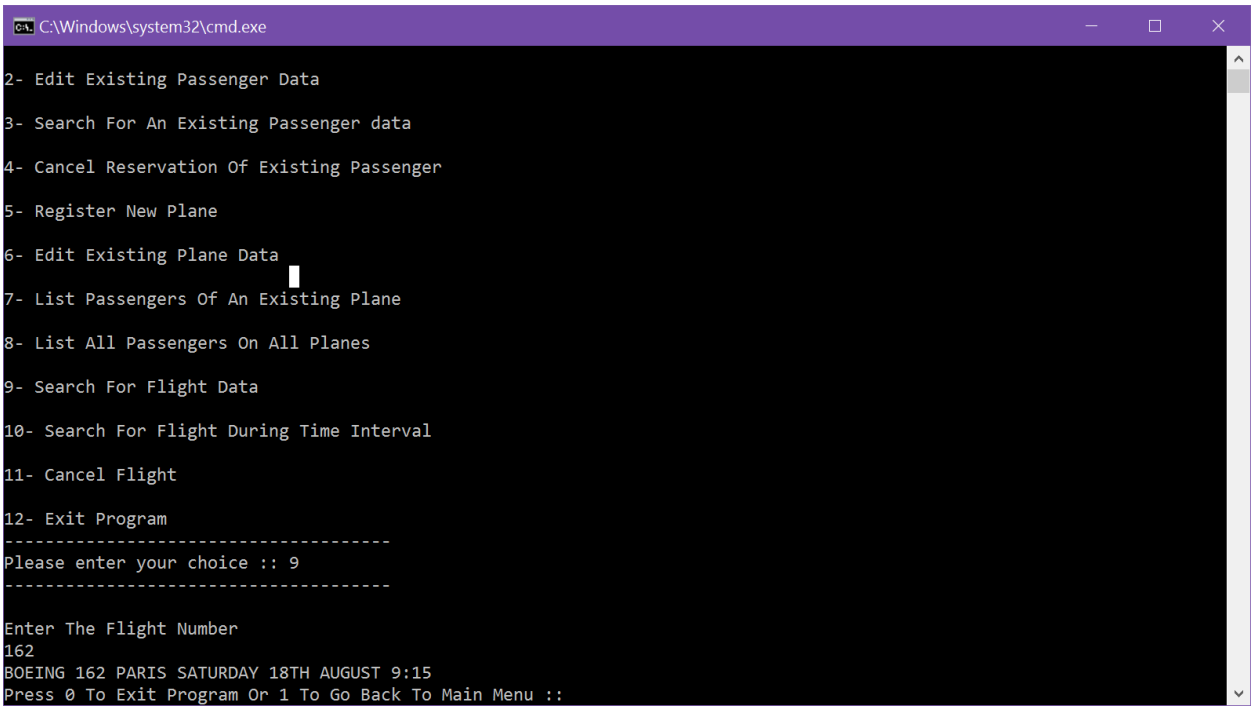


8. List of all passengers of all flights



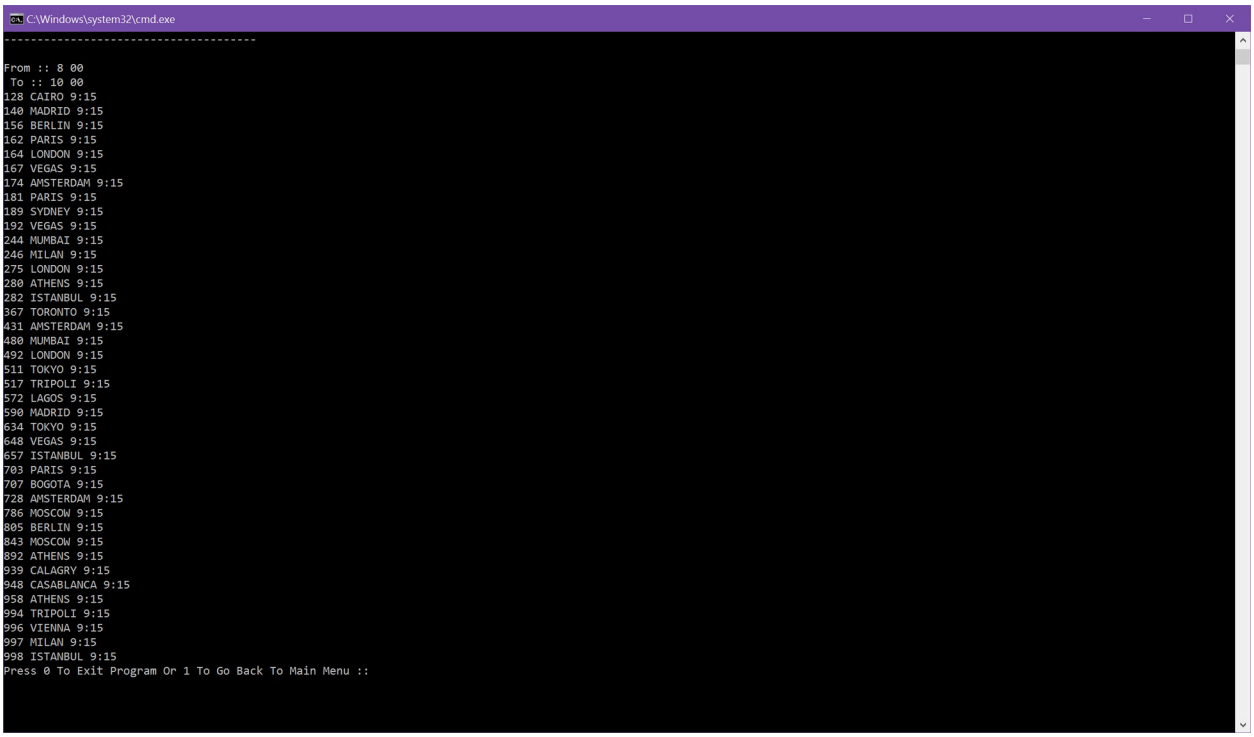
9. Search for flight data

Enter flight number. Flight's data will appear.



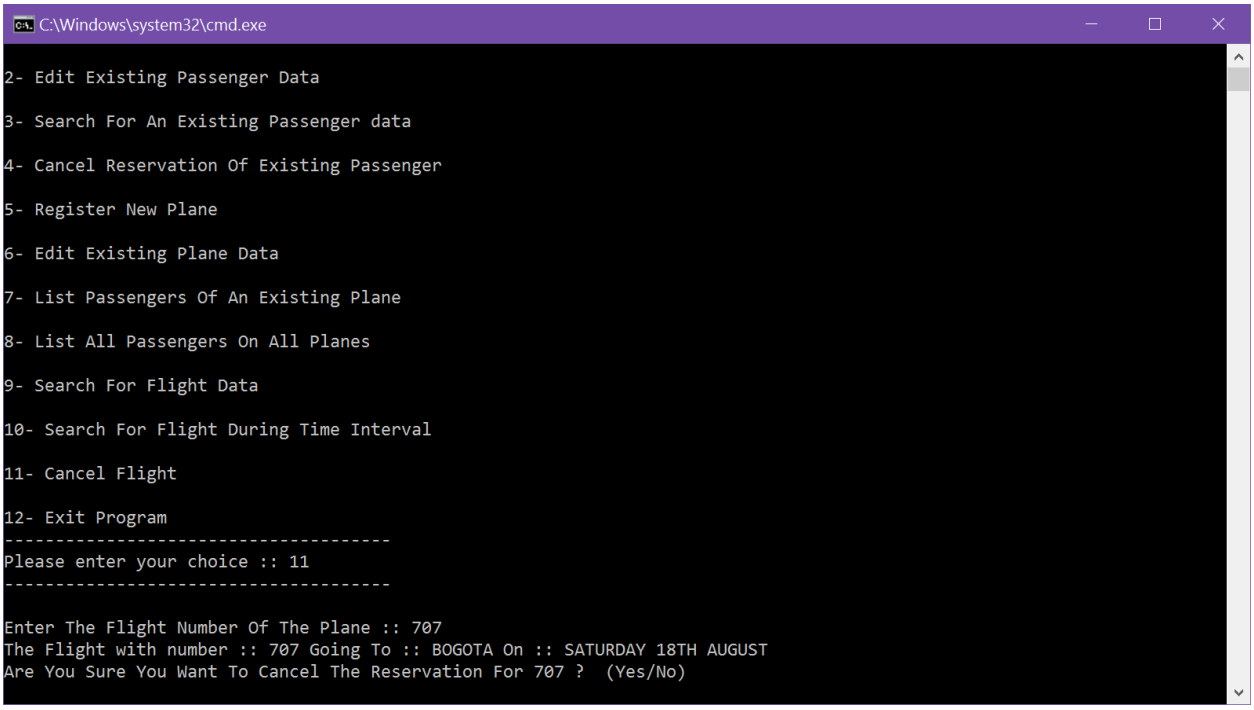
10. Search for flight during time interval

Enter time interval.

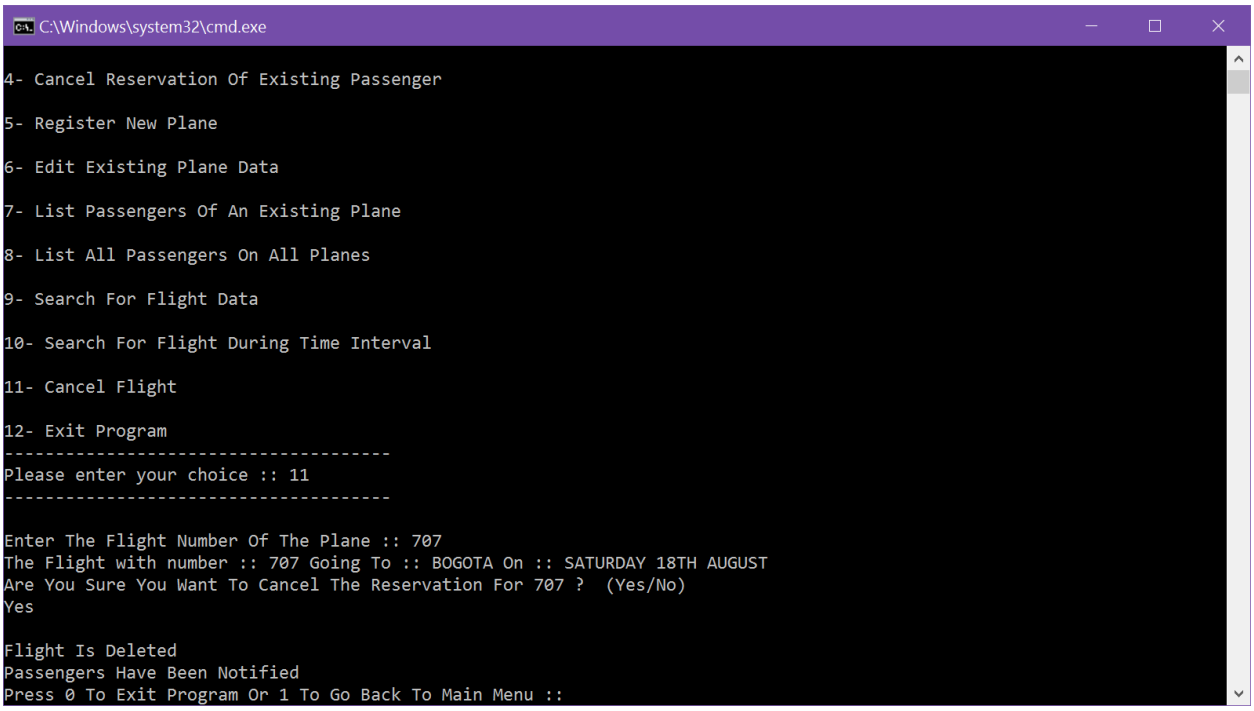


11. Cancel flights

Enter required flight number to be cancelled. Flight's data will appear. Type "Yes" if sure of this flight cancellation.

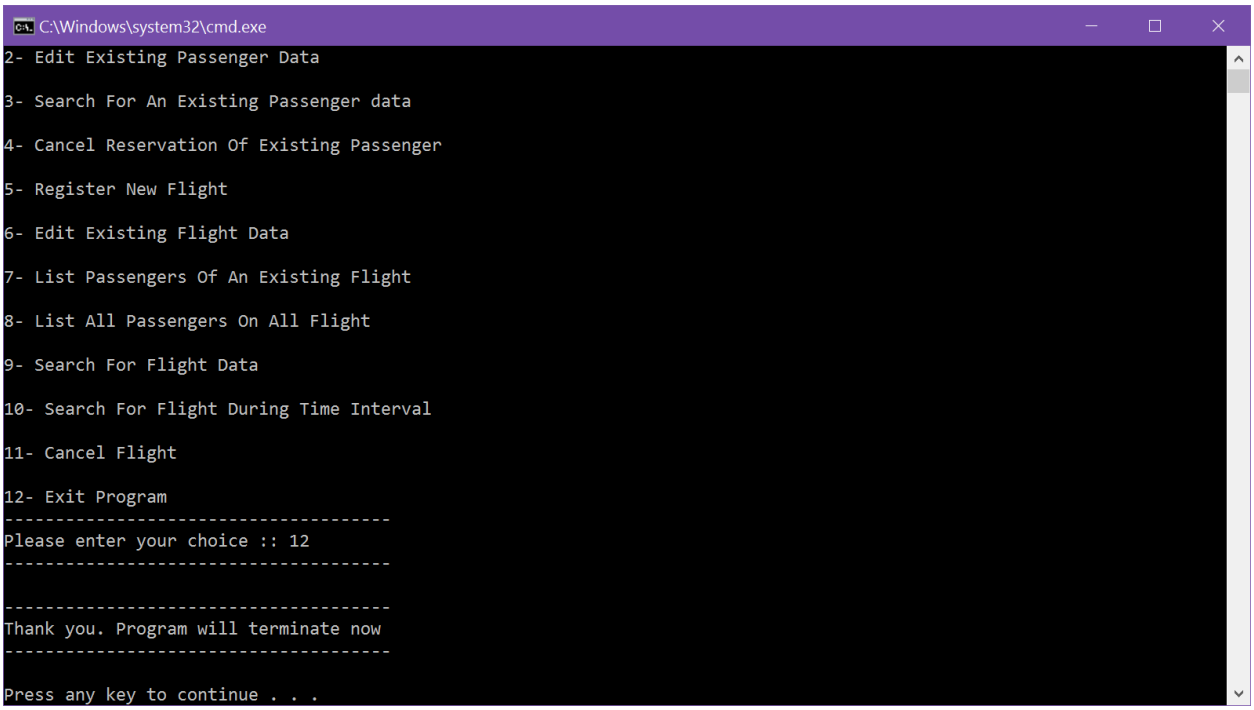


An announcement must appear to verify flight cancellation. Passengers will be notified.



12. Exit Program

Program terminates.



Error code list:

Error code	Description
31	Flight number already exist.
60	User didn't enter a correct operation choice from main menu.
71	User didn't enter 1 to go back to main menu or 0 to exit program. Program was forced to go back to main menu by default.
82	No passenger was found having this ID.
84	No flight was found having this flight no.
97	No flights were found during the entered time interval.

- Conclusion

This application allows airport employees to register new passengers, edit passengers' data, find passengers' data, cancel a passenger's reservation, register new planes, edit their data, show lists of passengers for every flight, search for flights, and cancel flights without any data corruption.

- References

Lafore, Robert. (2002). Object Oriented Programming in C++. USA: Sams publishing